PART 4 | EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis Report

A descriptive view of patient, clinical, operational, financial, and feedback records

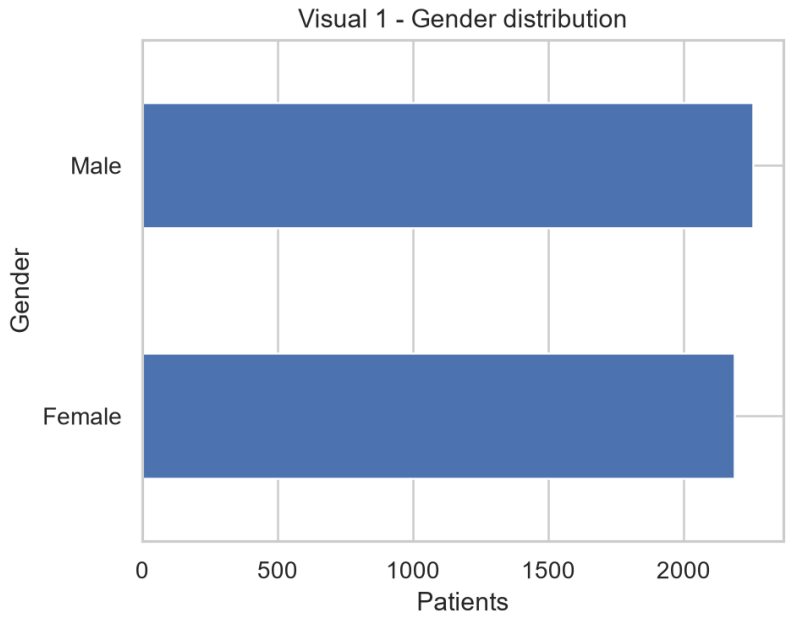
Executive readout

The appointment cohort contains 5,000 visits for 4,443 distinct patients. Recorded demographics and clinical events are broadly distributed across categories, while financial and experience measures show meaningful variation. Several commonly requested operational indicators cannot be measured directly because the source files contain rosters rather than time-stamped utilization or assignments.

Domain	Observed signal	Reading
Patient profile	Mean age 45.5 years; median 46; near-even gender split	Broad age span, not a population estimate.
Clinical activity	Diagnosis categories are relatively balanced; lab records link to 45.4% of visits	Counts show documentation/utilization, not incidence or test outcomes.
Operations	Doctor volumes 5-28 visits; all listed beds marked Available	Rosters are not workload or occupancy events.
Finance	Billed totals 57.2-70.2m per month; receipts remain lower	Source amount units; currency is unspecified.
Experience	1,756 rated feedback records; mean rating 2.93/5	Feedback covers 35.1% of appointments.

Patient and clinical profile

Age spans childhood through older adulthood, with the central half between 23 and 68 years. The largest age group in the engineered table is 65+, but repeated appointment rows mean this is an encounter-cohort count unless deduplicated to patients. Recorded chronic-condition categories are similar in size; 25.5% of distinct patients have no recorded history, limiting prevalence claims. Diagnosis counts do not show a single dominant category.

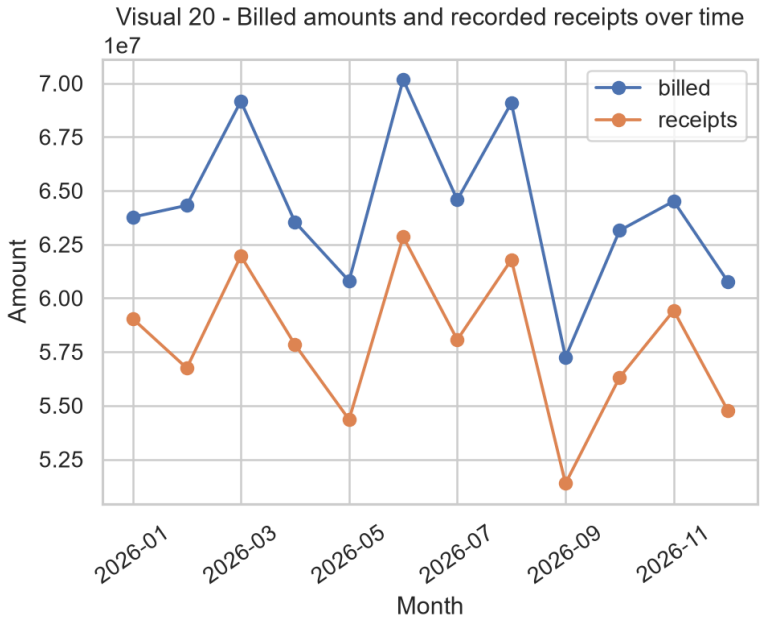


Visual 01 gender

Operational and financial patterns

Doctor appointment counts vary, but the source lacks shift duration, schedule, specialty case mix, and assignment history. The nurse and equipment rosters are nearly uniform by department. Every listed bed is labeled Available, which is a roster status rather than a time-based occupancy measure. Facility counts are relatively close among the highest-volume sites, but no capacity denominator supports utilization comparisons.

Linked billed amounts and recorded receipts generally move together over the observed months, with receipts below billed totals. The financial status fields are not reconciled to payments: the calculated balance remains positive for records labeled Paid, while Pending records can also have payments. Department receipts are attributed through appointment-to-billing links; the unlinked Revenue_Summary table is excluded from department allocation.

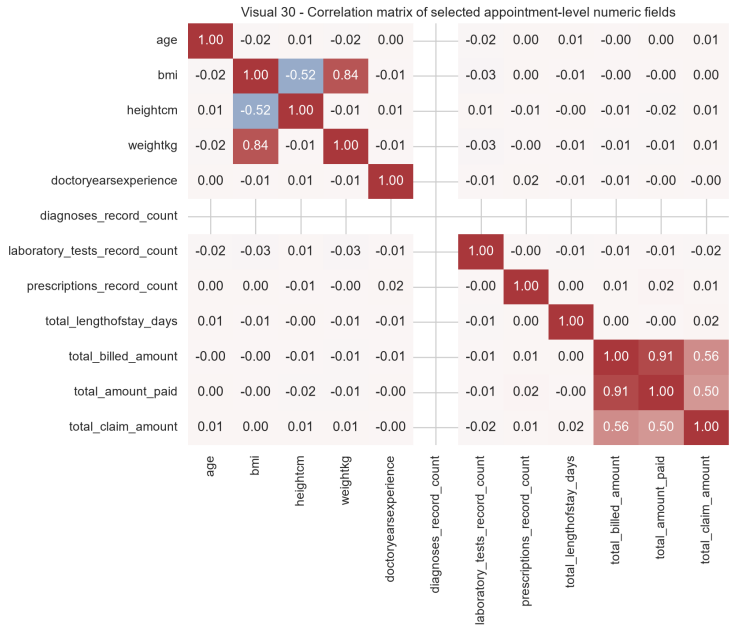


Visual 20 monthly billed and receipts

Patient experience and multivariate findings

Among 1,756 rated feedback records, 40.9% are ratings 1-2 and 37.0% are ratings 4-5; the mean is 2.93 and median is 3. Department means range from 2.76 to 3.02. The lowest and highest displayed facility averages differ by 0.43 points, but neither comparison adjusts for patient mix or response bias. Ratings should be treated as feedback from respondents, not a complete measure of service quality.

The selected numeric heatmap shows strong associations between BMI and weight (r = 0.84) and between billed and paid amount (r = 0.91). The latter partly reflects shared financial transactions. Most age, clinical-count, and length-of-stay relationships are weak. Visual inspection supports descriptive patterns but does not establish causes.



Visual 30 numeric correlation matrix

What the analysis supports

• Use the patient profile and linked encounter counts to plan further investigation, while keeping visit volume separate from population demand.

• Review low feedback scores and response coverage by department and facility before setting improvement targets.

• Treat monthly billed and receipt series as linked transactional indicators in unspecified source units, pending finance reconciliation.

• Collect dated staffing, bed, equipment, and facility-capacity data before using operational ratios as utilization measures.

Limitations

This is a retrospective analysis of the supplied records. The time series is short, no population denominators are supplied, some clinical histories are unrecorded, and feedback is incomplete. Geographic mapping and payment status inconsistencies remain unresolved. These limitations constrain generalization and operational ranking.